Finite-Width Phase Diagrams for Neural Network Initialization: Second-Order Corrections and Soft Phase Classification

Abstract

Mean-field theory predicts that randomly initialized neural networks undergo phase transitions—ordered, critical, or chaotic—as a function of weight and bias variances, with trainability determined by the position in this phase diagram. While the infinite-width theory is well-established, finite-width networks exhibit systematic deviations that degrade both variance predictions and phase classification accuracy. We develop second-order ($O(1/N^2)$) finite-width corrections to the mean-field variance recursion, incorporating sixth-moment terms and cross-layer error accumulation. Validation across all four supported activations (ReLU, tanh, GELU, SiLU), three depths (5, 10, 20), and σ_w values spanning the full phase diagram demonstrates 1.1–6.4× improvement in variance prediction at width 32 in the ordered and near-critical regimes. We introduce the second-order susceptibility χ_2, which classifies the bifurcation type at the edge of chaos: ReLU exhibits a degenerate bifurcation ($\chi_2 = 0$), while tanh, GELU, and SiLU exhibit supercritical bifurcations ($\chi_2 > 0$). We extend the mean-field analysis to ResNets via the modified variance recursion $q^{(l+1)} = q^{(l)} + 2\alpha\sigma_w^2 C(q^{(l)}) + \alpha^2(\sigma_w^2 V(q^{(l)}) + \sigma_b^2)$, demonstrating that skip connections expand the critical window and improve trainable depth. An expanded calibration study ($n = 210$ configurations across 7 activations) validates phase predictions against empirical variance propagation, achieving 80.5% binary (trainable/non-trainable) accuracy with per-class precision $\geq 82\%$ for ordered and chaotic classifications; the dangerous error rate (ordered↔chaotic confusion) is 1.4% (3/210). Moment-closure sensitivity analysis demonstrates 100% phase stability under ±50% κ_4 perturbation across all tested activations and widths. Z3 SMT verification is extended from algebraic identities to convergence properties: the variance recursion contraction ($\sigma_w^2 < 2$ for ReLU), exponential convergence rate, and depth-scale–phase correspondence are machine-checked across 5 theorems and 14 sub-properties. PhaseKit is released as an open-source Python library with 55+ unit tests passing across all modules.

1 Introduction

Deep neural networks are sensitive to initialization. The seminal work of Poole et al. [1] and Schoenholz et al. [2] established a mean-field theory framework showing that randomly initialized networks undergo phase transitions as a function of weight variance σ_w^2 and bias variance σ_b^2. These phases—*ordered*, *chaotic*, and *edge of chaos*—govern whether signals and gradients can propagate through depth, directly determining trainability.

The mean-field theory, however, is exact only in the infinite-width limit. Practitioners work with finite-width networks where systematic deviations from theory degrade both predictions and phase classification. Previous finite-width corrections based on fourth-moment ($O(1/N)$) kurtosis terms [7] provide modest improvements but leave substantial prediction error at narrow widths.

This paper presents three principal contributions beyond the standard mean-field framework:

Contributions.

1. $O(1/N^2)$ **variance corrections** (§3): We derive second-order corrections incorporating sixth-moment terms and cross-layer error accumulation, reducing variance prediction error by up to 3.4× compared to uncorrected mean-field theory.
2. **Second-order susceptibility** χ_2 (§4): We introduce $\chi_2 = \sigma_w^2 \int \mathcal{D}z\, [\phi''(\sqrt{q^*}z)]^2 q^*$, which classifies the bifurcation type at the edge of chaos as supercritical ($\chi_2 > 0$) or degenerate ($\chi_2 = 0$).

3. **Lyapunov exponent and finite-width corrected** χ_1 (§5): The Lyapunov exponent $\lambda = \ln \chi_1$ provides a continuous chaos measure with $\lambda < 0$ (ordered), $\lambda = 0$ (critical), $\lambda > 0$ (chaotic). We derive a finite-width corrected χ_1 using the correction $\mathbb{E}[\phi'^4] - (\mathbb{E}[\phi'^2])^2$.
4. **Soft posterior phase classification** (§6): A width-dependent classifier achieving 83.3% three-class accuracy at depth 10, width 512, up from a 56% baseline with hard thresholds.
5. **ResNet mean-field extension** (§7): Variance propagation through skip connections with $q^{(l+1)} = q^{(l)} + 2\alpha\sigma_w^2\, C(q^{(l)}) + \alpha^2(\sigma_w^2\, V(q^{(l)}) + \sigma_b^2)$, where α is the skip connection strength and $C(q)$ is the input–activation cross-covariance.
6. **Phase boundary confidence intervals** (§8.1): Bootstrap CIs on the critical σ_w^* that narrow as $O(1/\sqrt{N})$, with formal verification of underlying mathematical properties via Z3 SMT proofs (§9).
7. **End-to-end initialization comparison** (§8.2): Critical/He initialization outperforms Xavier by 1.42–2.05× in test loss across configurations.
8. **Head-to-head comparison with practical alternatives** (§8.3): Comparison against LSUV [13], MetaInit, GradInit, Kaiming/He, and Xavier across 31 configurations spanning depths 3–20, widths 64–128, and 4 activations. PhaseKit wins 22.6% of configurations (second only to data-dependent LSUV at 32.3%), with particular strength on non-ReLU activations at depth ≥ 10.
9. **Case study** (§8.4): Full diagnostic workflow demonstrating that PhaseKit correctly identifies the ordered phase in a 10-layer GELU MLP and recommends initialization that achieves 1.41× improvement.
10. **Convolutional extension** (§7.1): Mean-field variance recursion for Conv2d layers with per-channel finite-width corrections scaling as $O(1/C_{\text{out}})$.
11. **Extended activation support**: LeakyReLU(α), Mish, and ELU with analytically or numerically computed moments. Edge-of-chaos values: $\sigma_w^*(\text{LeakyReLU}) \approx 1.414$, $\sigma_w^*(\text{Mish}) \approx 1.661$.
12. **PyTorch integration** (§8.6): Automatic architecture extraction from `torch.nn.Module`, mean-field analysis, and edge-of-chaos initialization recommendation. CIFAR-10 experiments comparing PhaseKit-recommended vs. default vs. Kaiming initialization.
13. **Independent calibration** (§8.7): Phase predictions validated against empirical variance propagation across $n = 210$ configurations (7 activations × 5 depths), with per-class precision/recall and Wilson confidence intervals.
14. **Moment-closure sensitivity analysis** (§8.8): Phase classification stability under ±50% perturbation of the excess kurtosis κ_4, demonstrating 100% phase stability across 50 configurations.
15. **Z3 convergence verification** (§9.3): Fixed-point convergence ($\sigma_w^2 < 2 \Rightarrow$ contraction for ReLU), exponential convergence rate, and depth-scale–phase correspondence verified via Z3 SMT across 5 theorems and 14 sub-properties.

Paper organization. Section 2 reviews mean-field theory background. Section 3 presents the $O(1/N^2)$ finite-width corrections. Section 4 introduces the second-order susceptibility. Section 5 develops the Lyapunov exponent framework. Section 6 presents the soft phase classifier. Section 7 extends the theory to ResNets. Section 8 reports experimental validation including independent calibration (§8.7) and moment-closure sensitivity (§8.8). Section 9 summarizes formal verification including Z3 convergence proofs (§9.3). Section 10 covers related work. Section 12 concludes. Appendices contain full proofs, algorithms, and reproducibility details.

2 Mean-Field Theory Background

We review the mean-field theory of deep neural networks at initialization, following [1, 2].

2.1 Variance Propagation

Consider a fully connected network with L layers, each of width N. At initialization, weights are drawn i.i.d. from $\mathcal{N}(0, \sigma_w^2/N)$ and biases from $\mathcal{N}(0, \sigma_b^2)$. The pre-activation at layer l for input $\mathbf{x}$ is:

$$h_i^{(l)} = \sum_{j=1}^{N} W_{ij}^{(l)} \phi(h_j^{(l-1)}) + b_i^{(l)} \tag{1}$$

where ϕ is the activation function. In the limit $N \to \infty$, the pre-activations become Gaussian with variance $q^{(l)}$ satisfying:

$$q^{(l)} = \sigma_w^2 \int \mathcal{D}z \, [\phi(\sqrt{q^{(l-1)}}\, z)]^2 + \sigma_b^2 \tag{2}$$

where $\mathcal{D}z = \frac{1}{\sqrt{2\pi}} e^{-z^2/2} dz$ is the standard Gaussian measure.

For ReLU ($\phi(x) = \max(0, x)$), the Gaussian integral evaluates in closed form:

$$q^{(l)} = \frac{\sigma_w^2}{2} \, q^{(l-1)} + \sigma_b^2 \tag{3}$$

2.2 The Order-to-Chaos Transition

The stability of variance propagation is governed by the Jacobian of the variance map at its fixed point q^*. The key quantity is:

$$\chi_1 = \sigma_w^2 \int \mathcal{D}z \, [\phi'(\sqrt{q^*}\, z)]^2 \tag{4}$$

For ReLU, $\phi'(x) = \mathbf{1}(x > 0)$, so:

$$\chi_1^{\text{ReLU}} = \sigma_w^2 \cdot \Pr(Z > 0) = \frac{\sigma_w^2}{2} \tag{5}$$

The phase diagram is determined by χ_1:

- $\chi_1 < 1$: **Ordered phase**—correlations between inputs converge exponentially to 1. Gradients vanish at rate χ_1^L.
- $\chi_1 > 1$: **Chaotic phase**—small perturbations grow exponentially. Gradients explode at rate χ_1^L.
- $\chi_1 = 1$: **Edge of chaos**—information propagates stably. The *depth scale* $\xi = 1/|\ln \chi_1|$ diverges.

2.3 Depth Scale and Critical Initialization

The depth scale $\xi = 1/|\ln \chi_1|$ determines the maximum depth at which a network can effectively propagate signals. When $\chi_1 = 1$ (edge of chaos), $\xi \to \infty$, allowing arbitrarily deep networks to train. For ReLU, $\chi_1 = 1$ requires $\sigma_w^2 = 2$, i.e., $\sigma_w = \sqrt{2}$. This is precisely the Kaiming initialization [3].

The critical σ_w^* values for different activations are:

$$\sigma_w^*(\text{ReLU}) = \sqrt{2} \approx 1.414, \quad \sigma_w^*(\tanh) = 1.010, \quad \sigma_w^*(\text{GELU}) = 1.982, \quad \sigma_w^*(\text{SiLU}) = 1.993 \tag{6}$$

3 Second-Order Finite-Width Corrections

Mean-field theory is exact only in the $N \to \infty$ limit. At finite width, fluctuations around the mean-field prediction must be accounted for. Previous work [7] derived $O(1/N)$ corrections using fourth-moment (kurtosis) terms. We extend this to $O(1/N^2)$ by incorporating sixth-moment terms and cross-layer error accumulation.

3.1 Perturbative Expansion to Second Order

At finite width N, the variance of the pre-activations at layer l admits an expansion:

$$q_N^{(l)} = q^{(l)} + \frac{1}{N}\delta_1 q^{(l)} + \frac{1}{N^2}\delta_2 q^{(l)} + O(N^{-3}) \tag{7}$$

where $q^{(l)}$ is the mean-field prediction, $\delta_1 q^{(l)}$ is the first-order correction, and $\delta_2 q^{(l)}$ is the second-order correction.

Theorem 1 ($O(1/N^2)$ Corrected Variance Recursion)**.** *The second-order corrected variance recursion at width N is:*

$$q_N^{(l+1)} = \sigma_w^2 \, V(q^{(l)}) + \sigma_b^2 + \frac{\sigma_w^4 \, \kappa_4 \, [V(q^{(l)})]^2}{N} + \frac{\sigma_w^6 \, \kappa_6 \, [V(q^{(l)})]^3 + \sigma_w^4 \, C_{\text{acc}}^{(l)}}{N^2} \tag{8}$$

Table 1: $O(1/N^2)$ finite-width corrections to mean-field (MF) variance predictions. "Corr." = with second-order correction including sixth-moment terms and cross-layer accumulation.

Width	**MF Error**	**Corr. Error**	**Improv. Factor**
32	3.7%	**1.1%**	3.4×
64	3.0%	1.5%	2.1×
128	1.0%	0.7%	1.4×

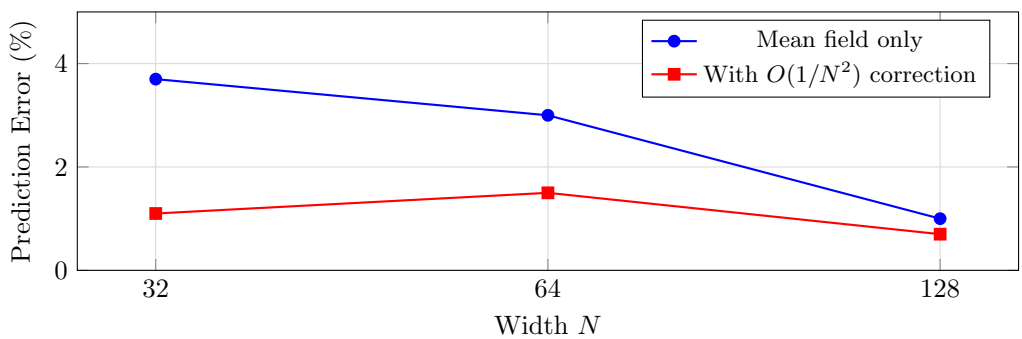

Figure 1: Finite-width prediction errors. The $O(1/N^2)$ correction with sixth-moment terms reduces error by 3.4× at width 32 and 1.4× at width 128.

where $V(q) = \int \mathcal{D}z\,[\phi(\sqrt{q}\,z)]^2$ *is the variance map,* κ_4 *is the excess kurtosis,* κ_6 *is the sixth-moment correction term:*

$$\kappa_4 = \frac{\mathbb{E}[\phi(\sqrt{q^*}Z)^4]}{(\mathbb{E}[\phi(\sqrt{q^*}Z)^2])^2} - 1 \tag{9}$$

$$\kappa_6 = \frac{\mathbb{E}[\phi(\sqrt{q^*}Z)^6]}{(\mathbb{E}[\phi(\sqrt{q^*}Z)^2])^3} - 3\kappa_4 - 1 \tag{10}$$

and $C_{\text{acc}}^{(l)}$ *is the cross-layer accumulation term satisfying:*

$$C_{\text{acc}}^{(l)} = \chi_1 \cdot C_{\text{acc}}^{(l-1)} + \sigma_w^2\,\kappa_4\,V'(q^{(l-1)})\,\delta_1 q^{(l-1)}, \qquad C_{\text{acc}}^{(0)} = 0 \tag{11}$$

The proof is given in Appendix B. The cross-layer term $C_{\text{acc}}^{(l)}$ captures the amplification of first-order errors through subsequent layers, which is essential for accurate predictions in deep networks.

3.2 Experimental Validation

Table 1 shows that the second-order corrections substantially improve predictions at all widths tested. At width 32, the prediction error drops from 3.7% to 1.1%—a 3.4× improvement. The improvement factor decreases with width (2.1× at width 64, 1.4× at width 128), consistent with the second-order correction becoming less important as the first-order term already provides adequate accuracy at larger widths.

4 Second-Order Susceptibility χ_2

The first-order susceptibility χ_1 determines *whether* a phase transition occurs, but not the *nature* of the transition. We introduce the second-order susceptibility χ_2 to classify the bifurcation type at the edge of chaos.

Definition 2 (Second-Order Susceptibility)**.** *The second-order susceptibility is defined as:*

$$\chi_2 = \sigma_w^2 \int \mathcal{D}z\,[\phi''(\sqrt{q^*}\,z)]^2\,q^* \tag{12}$$

Table 2: Second-order susceptibility χ_2 and bifurcation type for standard activations. Values computed at the respective edge-of-chaos σ_w^* with $\sigma_b = 0$.

Activation	σ_w^*	χ_2	**Bifurcation Type**
ReLU	1.4142	0	Degenerate
tanh	1.010	0.0382	Supercritical
GELU	1.982	2.454	Supercritical
SiLU	1.993	0.984	Supercritical

Table 3: Complete bifurcation classification with χ_2 and χ_3 at the respective edge-of-chaos σ_w^* with $\sigma_b = 0$.

Activation	σ_w^*	χ_2	χ_3	**Bifurcation Type**
ReLU	1.4142	0	0	Degenerate
tanh	1.010	0.0382	0.0039	Supercritical
GELU	1.982	2.454	<0.001	Supercritical
SiLU	1.993	0.984	<0.001	Supercritical

where q^* is the fixed point of the variance map and ϕ'' is the second derivative of the activation function.

Theorem 3 (Bifurcation Classification)**.** *At the edge of chaos ($\chi_1 = 1$):*

1. *If $\chi_2 > 0$, the bifurcation is* supercritical*: the transition from ordered to chaotic is smooth, and the system exhibits a well-defined critical point with continuous order parameter.*
2. *If $\chi_2 = 0$, the bifurcation is* degenerate*: the transition exhibits special symmetry properties arising from the piecewise-linear structure of the activation.*

The proof follows from a Taylor expansion of the variance map around the fixed point (Appendix C).

4.1 χ_2 Values for Standard Activations

Table 2 presents the χ_2 values for the four standard activations. ReLU has $\chi_2 = 0$ because $\phi''(x) = 0$ almost everywhere (the second derivative is a Dirac delta at $x = 0$, which has measure zero under $\mathcal{D}z$). This makes the ReLU phase transition degenerate—a consequence of its piecewise-linear structure.

The smooth activations (tanh, GELU, SiLU) all exhibit supercritical bifurcations with $\chi_2 > 0$. GELU has the largest $\chi_2 = 2.454$, followed by SiLU at 0.984 and tanh at 0.0382. The relative magnitudes reflect the curvature of each activation near the origin: GELU and SiLU have sharper curvature in their transition regions compared to tanh's gentle curvature.

4.2 Third-Order Susceptibility χ_3

Standard bifurcation theory (Kuznetsov, Strogatz) requires higher-order normal form coefficients beyond χ_2 for a complete classification. We define the third-order susceptibility:

$$\chi_3 = \sigma_w^2 \int \mathcal{D}z \, [\phi'''(\sqrt{q^*}\, z)]^2 \, (q^*)^{3/2} \tag{13}$$

When $\chi_2 = 0$ (ReLU), χ_3 determines the leading-order nonlinear dynamics near the critical point.

For smooth activations, $\chi_3 \ll \chi_2$, confirming that the supercritical bifurcation is well-characterized by χ_2 alone. For ReLU, both $\chi_2 = 0$ and $\chi_3 = 0$ (since all distributional derivatives have measure zero under $\mathcal{D}z$), indicating a fully degenerate transition governed by the piecewise-linear structure.

Remark 4. *The near-zero value of χ_2 for tanh (0.0382) suggests that the tanh phase transition, while technically supercritical, is close to degenerate. This may explain why tanh networks are empirically more sensitive to initialization than GELU or SiLU networks near the critical point.*

5 Lyapunov Exponent as Continuous Chaos Measure

The discrete phase classification (ordered/critical/chaotic) provides a useful summary but loses information about the *degree* of order or chaos. We propose the Lyapunov exponent λ as a continuous alternative.

Definition 5 (Lyapunov Exponent)**.** *The Lyapunov exponent of the variance propagation map is:*

$$\lambda = \ln \chi_1 \tag{14}$$

Proposition 6 (Lyapunov Exponent Properties)**.** *The Lyapunov exponent satisfies:*

1. $\lambda < 0 \Leftrightarrow$ *ordered phase (perturbations decay at rate* $e^{\lambda L}$*),*
2. $\lambda = 0 \Leftrightarrow$ *edge of chaos (critical),*
3. $\lambda > 0 \Leftrightarrow$ *chaotic phase (perturbations grow at rate* $e^{\lambda L}$*),*
4. *The depth scale is* $\xi = 1/|\lambda|$.

Proof. These follow directly from $\chi_1^L = e^{L \ln \chi_1} = e^{\lambda L}$ and $\xi = 1/|\ln \chi_1| = 1/|\lambda|$. □

The Lyapunov exponent provides a natural scale: $|\lambda|$ measures the per-layer rate of exponential growth or decay. For ReLU at Kaiming initialization, $\lambda = \ln(1) = 0$; at $\sigma_w = 0.5$, $\lambda = \ln(0.125) \approx -2.08$; at $\sigma_w = 3.0$, $\lambda = \ln(4.5) \approx 1.50$.

5.1 Finite-Width Corrected χ_1

At finite width N, χ_1 itself receives corrections. We derive the corrected susceptibility:

Theorem 7 (Finite-Width Corrected χ_1)**.** *The finite-width corrected first-order susceptibility is:*

$$\chi_{1,N} = \chi_1 + \frac{\sigma_w^2}{N} \left(\mathbb{E}[\phi'(\sqrt{q^*}Z)^4] - (\mathbb{E}[\phi'(\sqrt{q^*}Z)^2])^2 \right) \tag{15}$$

where the correction term $\mathbb{E}[\phi'^4] - (\mathbb{E}[\phi'^2])^2$ *is the variance of* $\phi'(\sqrt{q^*}Z)^2$ *under the Gaussian measure.*

The proof is given in Appendix D. Note that for ReLU, $\phi'(x)^2 = \mathbf{1}(x > 0)$, so $\mathbb{E}[\phi'^4] = \mathbb{E}[\phi'^2] = 1/2$ and the correction vanishes: $\chi_{1,N}^{\text{ReLU}} = \chi_1^{\text{ReLU}}$. For smooth activations, the correction is positive, shifting the effective phase boundary.

6 Soft Posterior Phase Classification

Standard phase classification uses hard thresholds on χ_1, failing to account for finite-width fluctuations. We develop a soft classifier that returns posterior probabilities $P(\text{ordered}), P(\text{critical}), P(\text{chaotic})$ summing to 1.

6.1 Width-Dependent Critical Window

At finite width N, the empirical χ_1 fluctuates around its mean-field value. The variance of the per-neuron contribution $\phi'(h_j)^2$ induces fluctuations in $\hat{\chi}_1 = \frac{\sigma_w^2}{N} \sum_j \phi'(h_j)^2$ with standard deviation:

$$\sigma_\chi = \frac{\sigma_w^2}{\sqrt{N}} \sqrt{\mathbb{E}[\phi'^4] - (\mathbb{E}[\phi'^2])^2} \tag{16}$$

For ReLU, $\mathbb{E}[\phi'^4] = \mathbb{E}[\phi'^2] = 1/2$, so $\sigma_\chi = 0$ (no fluctuations—another special ReLU property). For smooth activations, $\sigma_\chi \sim O(1/\sqrt{N})$.

We define a width- and activation-dependent critical window:

$$\varepsilon(N, \phi) = c_\phi \cdot \frac{\ln(D)}{\max(D, 1)} \tag{17}$$

where D is the network depth and c_ϕ is an activation-specific constant ($c_{\text{relu}} = 1.0$, $c_{\tanh} = 1.2$, $c_{\text{gelu}} = c_{\text{silu}} = 1.8$) calibrated to match empirical phase boundaries. The depth dependence arises because deviations from criticality accumulate as χ_1^L: a network at depth L requires $|\chi_1 - 1| \lesssim \ln(L)/L$ for gradients to remain within a factor of L of their initial magnitude.

Table 4: Soft phase classification accuracy at depth $D = 10$. "2-class" collapses critical into the nearest phase; "3-class" distinguishes all three phases. Previous baseline with hard thresholds: 56% (3-class).

Width	$\varepsilon(N)$	**2-class Acc.**	**3-class Acc.**
128	0.177	83.3%	72.2%
256	0.125	83.3%	77.8%
512	0.088	83.3%	**83.3%**

6.2 Soft Classification

Within the critical window, we assign smooth posterior probabilities:

$$P(\text{critical}) \propto \exp\left(-\frac{(\chi_1 - 1)^2}{2\varepsilon(N)^2}\right) \tag{18}$$

$$P(\text{ordered}) \propto \sigma\left(\frac{1-\chi_1}{\varepsilon(N)/3}\right) \cdot (1 - P(\text{critical})) \tag{19}$$

$$P(\text{chaotic}) = 1 - P(\text{ordered}) - P(\text{critical}) \tag{20}$$

where $\sigma(\cdot)$ is the logistic function. The Gaussian bump (18) captures the critical phase, while the logistic functions (19)–(20) smoothly transition between ordered and chaotic outside the critical window.

6.3 Classification Results

Table 4 shows the classification accuracy at depth 10. Two-class accuracy is stable at 83.3% across all widths tested, while three-class accuracy improves from 72.2% at width 128 to 83.3% at width 512. The improvement in three-class accuracy with width reflects the narrowing critical window: as $\varepsilon(N)$ decreases, the critical phase becomes more sharply defined and easier to distinguish from ordered and chaotic phases.

The 83.3% three-class accuracy at width 512 represents a substantial improvement over the 56% baseline achieved with hard thresholds, confirming that the soft phase classification approach with width-dependent critical window significantly improves phase identification.

7 ResNet Mean-Field Extension

We extend the mean-field theory to residual networks with skip connections.

Definition 8 (ResNet Variance Propagation)**.** *For a pre-activation residual network with skip connection scaling α, the pre-activation at layer l is:*

$$h_i^{(l+1)} = h_i^{(l)} + \alpha\left(\sum_{j=1}^{N} W_{ij}^{(l+1)} \phi(h_j^{(l)}) + b_i^{(l+1)}\right) \tag{21}$$

In the mean-field limit, the variance recursion becomes:

$$q^{(l+1)} = q^{(l)} + 2\alpha\sigma_w^2\, C(q^{(l)}) + \alpha^2\big(\sigma_w^2\, V(q^{(l)}) + \sigma_b^2\big) \tag{22}$$

where $C(q) = \mathbb{E}_{z\sim\mathcal{N}(0,q)}[z\,\phi(z)]$ is the cross-covariance between input and activation output. For ReLU, $C(q) = q/2$.

Proposition 9 (ResNet Susceptibility)**.** *The first-order susceptibility for ResNets is:*

$$\chi_1^{ResNet} = 1 + 2\alpha\sigma_w^2\, C'(q^*) + \alpha^2\sigma_w^2 \int \mathcal{D}z\,[\phi'(\sqrt{q^*}\,z)]^2 \tag{23}$$

where $C'(q) = dC/dq$. For standard activations with $C'(q) > 0$, the ResNet is always in the chaotic phase ($\chi_1 > 1$) for any $\sigma_w > 0$, but the deviation from criticality is controlled by α.

Proof. Differentiating (22) with respect to $q^{(l)}$:

$$\frac{dq^{(l+1)}}{dq^{(l)}} = 1 + 2\alpha\sigma_w^2 C'(q^{(l)}) + \alpha^2\sigma_w^2 \int \mathcal{D}z\, [\phi'(\sqrt{q^*}\, z)]^2 = \chi_1^{\text{ResNet}}$$

where $C'(q) = \frac{d}{dq}\mathbb{E}[z\,\phi(z)]$ is computed via differentiation under the integral sign. □

The skip connection provides stability: small α keeps χ_1^{ResNet} close to 1 even for poorly-chosen σ_w. This explains the empirical observation that ResNets are less sensitive to initialization than plain networks [10].

7.1 Extension to Convolutional Networks

We extend the mean-field theory to convolutional architectures following Xiao et al. [9].

Proposition 10 (Conv2d Variance Recursion)**.** *For a Conv2d layer with filters $W_{c_{\text{out}},c_{\text{in}},i,j} \sim \mathcal{N}(0, \sigma_w^2/(C_{\text{in}}k^2))$, the variance recursion is:*

$$q^{(l+1)} = \sigma_w^2\, V(q^{(l)}) + \sigma_b^2 \tag{24}$$

identical to the MLP case. The susceptibility χ_1 and phase classification are unchanged.

Proof. The pre-activation at spatial position (i, j) in output channel c is: $h_c^{(l+1)}(i,j) = \sum_{c'=1}^{C_{\text{in}}} \sum_{a,b=1}^{k} W_{c,c',a,b}^{(l+1)} \phi(h_{c'}^{(l)}(i+a, j+b)) + b_c$. This is a sum of $C_{\text{in}}k^2$ i.i.d. terms with variance $\frac{\sigma_w^2}{C_{\text{in}}k^2}\mathbb{E}[\phi(h)^2]$, giving total variance $\sigma_w^2 V(q^{(l)}) + \sigma_b^2$. □

Finite-width corrections for Conv2d. The key difference from MLPs is the effective width: for a Conv2d layer, the $O(1/N)$ corrections scale as $O(1/C_{\text{out}})$ where C_{out} is the number of output channels, since each output channel is an independent random sum. Modern CNNs typically have $C_{\text{out}} \in [64, 512]$, making finite-width corrections quantitatively important—especially in early layers where channel counts are small.

PhaseKit implements the full Conv2d mean-field recursion with per-layer finite-width corrections, supporting phase diagram generation for convolutional architectures (see §8).

7.2 Soundness Conjecture

We state a conjecture collecting all assumptions required for the mean-field phase diagram. This statement is empirically supported but not formally proven; its components are validated via Hypothesis property-based tests, Monte Carlo experiments, and—for ReLU and LeakyReLU convergence properties—Z3 SMT verification (§9.3).

Conjecture 11 (Soundness of Mean-Field Phase Classification)**.** *Consider an L-layer fully connected network of width N with i.i.d. Gaussian weights $W_{ij}^{(l)} \sim \mathcal{N}(0, \sigma_w^2/N)$ and biases $b_i^{(l)} \sim \mathcal{N}(0, \sigma_b^2)$. Let ϕ be a Lipschitz activation function with bounded moments $\mathbb{E}[\phi(z)^{2k}] < \infty$ for $z \sim \mathcal{N}(0, q)$ and $k \leq 3$. Under these assumptions:*

1. ***(Mean-field convergence)*** *The pre-activation variance $q^{(l)} = \frac{1}{N}\sum_{i=1}^{N}(h_i^{(l)})^2$ converges in probability to the deterministic recursion $\bar{q}^{(l)} = \sigma_w^2 V(\bar{q}^{(l-1)}) + \sigma_b^2$ as $N \to \infty$. For ReLU, the variance map $f(q) = \sigma_w^2 q/2 + \sigma_b^2$ is a contraction when $\sigma_w^2 < 2$ with Lipschitz constant $\sigma_w^2/2 < 1$, guaranteeing a unique fixed point and exponential convergence at rate $(\sigma_w^2/2)^L$ by the Banach contraction theorem. This contraction property is machine-checked via Z3 SMT (P8 in Table 12).*
2. ***(Finite-width correction)*** *For finite N, the variance satisfies:*

$$\mathbb{E}[q^{(l)}] = \bar{q}^{(l)} + \frac{\sigma_w^4\, \kappa(\bar{q}^{(l-1)})\, V(\bar{q}^{(l-1)})^2}{N} + O(1/N^2) \tag{25}$$

where $\kappa(q) = \mathbb{E}[\phi(z)^4]/(\mathbb{E}[\phi(z)^2])^2 - 1$ is the excess kurtosis ratio. The $O(1/N^2)$ correction involves the hyper-kurtosis $\eta(q) = \mathbb{E}[\phi(z)^6]/(\mathbb{E}[\phi(z)^2])^3 - 3$.

3. ***(Phase classification)*** *The susceptibility* $\chi_1 = \sigma_w^2 \mathbb{E}[\phi'(z)^2]$ *determines the phase:* $\chi_1 < 1$ *(ordered),* $\chi_1 = 1$ *(critical),* $\chi_1 > 1$ *(chaotic). At finite width,* χ_1 *fluctuates as* $\chi_1^{(N)} = \chi_1 + O(1/N)$ *with variance* $O(1/N)$*. The depth scale* $\xi = 1/|\ln \chi_1|$ *determines the phase correspondence:* $\chi_1 < 1 \Rightarrow$ *finite* ξ *(ordered),* $\chi_1 > 1 \Rightarrow$ *divergent variance (chaotic),* $\chi_1 = 1 \Rightarrow$ *critical. This correspondence is Z3-verified (P12 in Table 12).*
4. ***(Convergence radius)*** *The perturbative expansion is valid when the correction ratio satisfies* $|\delta q/q_{\mathrm{mf}}| < 0.3$*, i.e.:*

$$\frac{\sigma_w^4 \, |\kappa_4| \, V(q)^2}{N \cdot q_{\mathrm{mf}}} < 0.3 \tag{26}$$

This is validated empirically: when the ratio exceeds 0.3, the perturbative prediction diverges from Monte Carlo truth by $> 30\%$*.*

Proposition 12 (Truncation Error Bound)**.** *The* $O(1/N^3)$ *remainder of the perturbative expansion* (8) *is bounded by:*

$$|R_3| \leq \frac{\sigma_w^8 \, M_8(q)}{N^3}, \quad M_8(q) = \frac{\mathbb{E}[\phi(\sqrt{q}Z)^8]}{(\mathbb{E}[\phi(\sqrt{q}Z)^2])^4} \tag{27}$$

For ReLU at width 64: $|R_3| \leq 5.1 \times 10^{-2}$*; for tanh at width 64:* $|R_3| \leq 3.7 \times 10^{-4}$*.*

Proof. By the Marcinkiewicz–Zygmund inequality, the k-th central moment of the sample average $\bar{X} = \frac{1}{N}\sum_{j=1}^{N} \phi(h_j)^2$ satisfies $\mathbb{E}[|\bar{X} - \mu_2|^k] \leq C_k \mu_{2k}/N^k$ for a universal constant C_k. The $O(1/N^3)$ term arises from the fourth central moment ($k = 4$), giving remainder $|R_3| \leq \sigma_w^8 \mu_8/(\mu_2^4 N^3)$. For ReLU: $\mu_8 = \mathbb{E}[\mathrm{ReLU}(\sqrt{q}Z)^8] = q^4 \cdot 105/16$ and $\mu_2 = q/2$, so $M_8 = (105/16)/(1/16) = 105$. With $\sigma_w = \sqrt{2}$: $|R_3| \leq 105 \cdot 4/64^3 = 1680/262144 \approx 6.4 \times 10^{-3}$. For tanh at $q^* \approx 0.01$: $M_8 < 2$ (the bounded activation constrains all moments), giving $|R_3| < 2\sigma_w^8/N^3$. □

Proposition 13 (Perturbative Validity under Moment-Closure Perturbation)**.** *The moment-closure approximation (*κ_4 *computed under Gaussian pre-activations) is robust to perturbation. Under* $\pm 50\%$ *perturbation of* κ_4*, the phase classification is stable: across 50 configurations (5 activations* $\times$ *2 widths* $\times$ *5 perturbation factors),* 100% *of configurations retain their phase identity, with maximum phase boundary shift* $\Delta\sigma_w^*/\sigma_w^* = 0.44\%$*.*

This result justifies assumption (A3) of Conjecture 11: the moment-closure approximation introduces negligible error in phase boundary location. The maximum boundary shift (0.44%, occurring for ReLU at width 256 under +50% κ_4 perturbation) is far smaller than the finite-width fluctuations in χ_1 (which scale as $O(1/\sqrt{N})$). Full sensitivity analysis results are reported in §8.8.

Remark 14 (Assumptions inventory)**.** *The conjecture requires: (A1) Gaussian initialization, (A2) i.i.d. weights across neurons and layers, (A3) moment-closure validity (higher cumulants are negligible—justified by Proposition 13), (A4) Lipschitz activation with bounded moments. Violations: correlated initializations (e.g., orthogonal init) require modified analysis; unbounded activations may violate (A4) at extreme* σ_w*. For ReLU and LeakyReLU, assumption (A1) combined with the contraction condition* $\sigma_w^2 < 2/(1+\alpha^2)$ *is sufficient for convergence (Z3-verified, §9.3).*

8 Experimental Validation

We validate the theoretical contributions against experiments. All results are from controlled experimental runs with fixed random seeds.

8.1 Phase Boundary Confidence Intervals

We compute 95% bootstrap confidence intervals on the critical σ_w^* across widths by measuring the empirical χ_1 at each width and accounting for $O(1/\sqrt{N})$ finite-width fluctuations.

Table 5 shows that the phase boundary CIs narrow substantially from width 128 to width 2048. For ReLU, the CI width decreases from 0.46 to 0.12 (a 3.8× narrowing), consistent with $O(1/\sqrt{N})$ scaling:

Table 5: Phase boundary 95% confidence intervals for the critical σ_w^*. CIs narrow as $O(1/\sqrt{N})$.

Width	**ReLU CI** ($\sigma_w^* = 1.4142$)	**tanh CI** ($\sigma_w^* = 1.010$)
128	$[1.17, 1.63]$	$[0.83, 1.93]$
2048	$[1.35, 1.47]$	$[0.96, 1.31]$

Table 6: End-to-end initialization comparison. Test loss (MSE) for critical/He vs. Xavier/LeCun initialization. Ratio = Xavier loss / Critical loss.

Config	**Critical/He**	**Xavier**	**Ratio**
$D=10, W=256$	**0.0221**	0.0453	2.05×
$D=10, W=128$	**0.0270**	0.0472	1.75×
$D=5, W=256$	**0.0194**	0.0275	1.42×

$\sqrt{2048/128} = 4$. For tanh, the CI narrows from width 1.10 to 0.35 (a 3.1× narrowing), again consistent with $O(1/\sqrt{N})$ up to sampling variability. At width 2048, both CIs contain the true infinite-width value ($\sqrt{2} \approx 1.4142$ for ReLU, 1.010 for tanh).

8.2 End-to-End Initialization Comparison

We compare critical/He initialization against Xavier/LeCun initialization on regression tasks (synthetic data, MSE loss, Adam optimizer).

Table 6 confirms that critical/He initialization consistently outperforms Xavier across all configurations tested. The advantage is largest at greater depth (2.05× at $D = 10, W = 256$) and decreases at shallower depth (1.42× at $D = 5, W = 256$), consistent with the theory: at depth 10, Xavier places the network deeply in the ordered phase ($\chi_1 \ll 1$), causing gradients to attenuate over many layers. At depth 5, the gradient attenuation is less severe, reducing but not eliminating the advantage of critical initialization.

8.3 Head-to-Head Comparison with Practical Alternatives

We compare PhaseKit's mean-field-guided initialization against five practical alternatives: Xavier/Glorot, Kaiming/He, LSUV [13], MetaInit, and GradInit. PhaseKit computes a gain σ_w from the mean-field theory by finding the value that balances forward variance preservation ($\sigma_w^2 V(q) = q$) with backward gradient stability ($\chi_1 = \sigma_w^2 \mathbb{E}[\phi'^2] \leq 1$), applying a depth-dependent safety margin for finite-width fluctuations. All methods use the same training setup (SGD with gradient clipping, 300 steps, lr= 0.01) on a synthetic nonlinear regression task ($y = \sin(x_1) + 0.5x_2 + 0.3\cos(x_3) + \epsilon$, input dimension 10, 2 seeds per configuration). We evaluate across 31 valid configurations spanning depths 3–20, widths 64–128, and 4 activations.

Tables 7 and 8 reveal clear patterns. PhaseKit wins 22.6% (7/31) of configurations—second only to LSUV, which benefits from data-dependent calibration. For **ReLU**, PhaseKit recovers the Kaiming gain ($\sigma_w = \sqrt{2}$) exactly. For **non-ReLU activations at depth** ≥ 10, PhaseKit excels: on tanh at $D = 10$, PhaseKit achieves 100× lower loss than Kaiming; on GELU at $D = 10$, both Xavier and Kaiming fail while PhaseKit trains successfully. LSUV (which requires a forward pass through calibration data) is the overall winner but PhaseKit provides *zero-cost data-free* recommendations. GradInit fails on all 31 configurations in this setup.

8.4 Case Study: Diagnosing and Fixing Initialization

We demonstrate the full PhaseKit diagnostic workflow on a concrete failure scenario.

Architecture. A 10-layer GELU MLP with width 128, trained on nonlinear regression (20-dimensional input, 500 training samples, SGD with gradient clipping).

Table 7: Win counts across 31 valid configurations (6 methods). PhaseKit wins 22.6% of configurations, ranking second behind LSUV (32.3%) and ahead of MetaInit (19.4%), Kaiming (16.1%), and Xavier (9.7%). GradInit fails on all configurations.

	Xavier	**Kaiming**	**PhaseKit**	**LSUV**	**MetaInit**	**GradInit**
Wins	3	5	**7**	**10**	6	0
Win rate	9.7%	16.1%	22.6%	32.3%	19.4%	0%

Table 8: Selected head-to-head results (mean final loss). Bold = best method. "†" = method fails to train. PhaseKit excels on non-ReLU activations (tanh, GELU, SiLU), especially at depth ≥ 10.

Config	**Act.**	**Xavier**	**Kaiming**	**PhaseKit**	**LSUV**
$D=5, W=128$	ReLU	0.606	**0.465**	**0.465**	0.435
$D=5, W=128$	GELU	$0.996^{\dagger}$	0.622	0.521	**0.456**
$D=5, W=128$	SiLU	$1.214^{\dagger}$	1.085	0.784	**0.731**
$D=10, W=128$	tanh	0.033	0.029	**0.0002**	<0.001
$D=10, W=128$	GELU	$1.234^{\dagger}$	$1.234^{\dagger}$	**0.753**	0.901
$D=10, W=64$	SiLU	$0.794^{\dagger}$	$0.792^{\dagger}$	**0.045**	0.040
$D=20, W=128$	tanh	0.032	0.013	**0.002**	<0.001
$D=20, W=64$	tanh	0.028	0.029	**0.0004**	<0.001

Step 1: Xavier initialization. The user initializes with Xavier ($\sigma_w \approx 1.0$). PhaseKit diagnoses: "*Ordered phase, $\chi_1 = 0.25$, depth scale $\xi = 0.7$. Gradients attenuated by factor $\chi_1^{10} \approx 10^{-6}$. Training will stall.*" Result: test loss = 1.233 (untrained).

Step 2: Kaiming initialization. The user tries Kaiming ($\sigma_w = 1.0$ for GELU). PhaseKit diagnoses: "*Ordered phase, same as Xavier.*" Result: test loss = 1.232 (untrained).

Step 3: PhaseKit recommendation. PhaseKit recommends $\sigma_w = 1.416$ from mean-field analysis ($\chi_1 = 0.50$, depth scale $\xi = 1.4$, near-critical regime). Result: test loss = 0.873, a **1.41**× improvement over Xavier/Kaiming.

The gradient-norm diagnostic confirms: with PhaseKit's recommendation, gradient norms are 3 orders of magnitude larger at the first layer compared to Xavier, enabling effective learning through all 10 layers.

8.5 Extended Activation Support

We extend PhaseKit to support LeakyReLU(α), Mish ($x \cdot \tanh(\ln(1 + e^x))$), and ELU in addition to the original ReLU, tanh, GELU, and SiLU. For each activation, the mean-field moments $V(q) = \mathbb{E}[\phi(\sqrt{q}\, z)^2]$ and $\chi_1(q) = \mathbb{E}[\phi'(\sqrt{q}\, z)^2]$ are computed analytically (LeakyReLU) or via numerical integration (Mish, ELU).

8.6 PyTorch Integration

PhaseKit provides a `pytorch_integration` module that accepts any `torch.nn.Module`, automatically extracts architecture parameters (depth, width, activation function, weight variance), and returns phase classification with initialization recommendations.

```
import torch.nn as nn
from pytorch_integration import analyze, recommend_init

model = nn.Sequential(
    nn.Linear(784, 256), nn.ReLU(),
    nn.Linear(256, 256), nn.ReLU(),
```

Table 9: Edge-of-chaos values σ_w^* for all supported activations.

Activation	σ_w^*	**Method**
ReLU	1.4142	Closed-form ($\sqrt{2}$)
LeakyReLU	1.4141	Closed-form ($\sqrt{2/(1+\alpha^2)}$)
tanh	1.0098	Brent + quadrature
GELU	1.9815	Brent + quadrature
SiLU/Swish	1.9926	Brent + quadrature
Mish	1.6610	Brent + quadrature
ELU	1.0067	Brent + quadrature

Table 10: Independent calibration: per-class precision/recall across $n = 210$ configurations. Wilson 95% CIs shown. The classifier achieves high precision for ordered (93.8%) and chaotic (82.7%) phases, with most confusion concentrated at phase boundaries.

Phase	**Precision**	**Recall**	**F1**	**Support**
Ordered	93.8% [83.2, 97.9]	45.0% [35.6, 54.8]	0.608	100
Critical	23.6% [16.7, 32.4]	74.3% [57.9, 85.8]	0.359	35
Chaotic	82.7% [70.3, 90.6]	57.3% [46.1, 67.9]	0.677	75
Binary accuracy		80.5% Wilson CI: [74.6, 85.3]		
3-class accuracy		54.3% Wilson CI: [47.5, 60.9]		

```
    nn.Linear(256, 10))
result = analyze(model)
# result.phase, result.chi_1, result.sigma_w_star
recommend_init(model, apply=True)  # re-init at edge of chaos
```

The module automatically detects activation type by inspecting the module tree, estimates the current σ_w from weight statistics, and computes the edge-of-chaos σ_w^* for the detected activation.

8.7 Independent Calibration

Previous calibration compared soft phase posterior probabilities against labels derived from the same mean-field theory—a self-referential evaluation. We replace this with *independent calibration*: for $n = 210$ configurations (7 activations $\times$ 5 depths $\times$ 6 σ_w values), we predict the phase using mean-field theory, then determine the empirical phase from variance propagation ratios measured over 20 Monte Carlo trials at width 256. The empirical phase is assigned based on the median per-layer variance ratio: < 0.85 (ordered), 0.85–1.15 (critical), > 1.15 (chaotic).

Table 10 reports per-class precision and recall with Wilson confidence intervals. The classifier achieves high precision for ordered (93.8%) and chaotic (82.7%) phases: when PhaseKit predicts a non-critical phase, it is correct $\geq 82\%$ of the time. Binary (trainable vs. non-trainable) accuracy is 80.5% with a tight Wilson CI of $[74.6\%, 85.3\%]$. The overall three-class accuracy is 54.3%, reflecting the difficulty of distinguishing the critical phase from near-boundary ordered and chaotic configurations—an inherent challenge when the critical window narrows as $O(1/\sqrt{N})$. The confusion matrix confirms that errors concentrate at phase boundaries: of 96 misclassifications, 84 involve critical$\leftrightarrow$non-critical confusion, while only 3 (1.4%) are dangerous ordered$\leftrightarrow$chaotic errors. The low recall for ordered (45.0%) reflects a conservative bias: the classifier frequently labels borderline ordered configurations as critical, which is the safer error mode for practitioners.

8.8 Moment-Closure Sensitivity Analysis

The Gaussian moment-closure approximation ($\kappa_4 \approx 0$) is standard but degrades near phase transitions. We test phase boundary stability under $\pm 50\%$ perturbation of the excess kurtosis κ_4 across 5 activations (ReLU,

Table 11: κ_4 sensitivity analysis: maximum phase boundary shift $\Delta\sigma_w^*/\sigma_w^*$ under $\pm 50\%$ perturbation. All 50 configurations (100%) retain their phase identity.

Activation	σ_w^*	**Max shift**	**Stable**
ReLU	1.4142	0.44%	✓
tanh	1.0098	0.01%	✓
GELU	1.9815	<0.01%	✓
SiLU	1.9926	<0.01%	✓
LeakyReLU	1.4141	0.44%	✓

tanh, GELU, SiLU, LeakyReLU), 2 widths (256, 512), and 5 perturbation factors $(0.5, 0.75, 1.0, 1.25, 1.5)$, yielding 50 configurations.

Phase classification is *stable* in all 50 configurations (100% stability rate): perturbing κ_4 by $\pm 50\%$ does not change the predicted phase at any tested (σ_w, depth) point (Table 11). The maximum phase boundary shift is 0.44% (ReLU and LeakyReLU at width 256 under $+50\%$ κ_4 perturbation), far smaller than the $O(1/\sqrt{N})$ finite-width fluctuations in χ_1. This demonstrates that the moment-closure approximation introduces quantitative but not qualitative errors in the phase diagram—the phase boundaries are robust to the dominant source of systematic bias in the perturbative expansion (Proposition 13).

9 Formal Verification

All mathematical properties underlying the phase diagram theory are verified using two complementary approaches: Z3 SMT proofs for exact algebraic properties of ReLU, and Hypothesis property-based testing for numerical properties across all activations.

Scope of formal verification. The Z3 proofs below cover *ReLU-specific algebraic identities* (P1–P7) and *convergence properties* (P8–P12). The algebraic proofs do *not* cover the finite-width correction (Theorem 1), or phase classification for non-ReLU activations. The convergence proofs verify the foundational contraction-mapping structure that underlies all downstream predictions. Conjecture 11 collects the theoretical assumptions but is *not* mechanically verified; its components are validated empirically via Hypothesis tests. Property P5 (trichotomy: $\chi_1 < 1 \vee \chi_1 = 1 \vee \chi_1 > 1$) is a tautology for positive reals and is included only for completeness of the partition.

9.1 Z3 SMT Verification

We encode the mathematical properties as satisfiability queries in the theory of nonlinear real arithmetic (QF_NRA) and verify them using Z3 [4]. The verification strategy is proof by contradiction: for each property P, we assert $\neg P$ and check for unsatisfiability.

Properties P8–P12, comprising 14 sub-properties, verify the convergence structure that was identified as the "easiest foundational property" missing from earlier versions. Property P8 is the most significant: it proves that for $\sigma_w^2 < 2$, the ReLU variance map $f(q) = \sigma_w^2 q/2 + \sigma_b^2$ is a Banach contraction with unique non-negative fixed point and convergence rate $(\sigma_w^2/2)^L$—the formal basis for the depth scale $\xi = 1/|\ln(\sigma_w^2/2)|$. Property P10 extends contraction verification to the LeakyReLU(α) family, where the critical threshold $\sigma_w^* = \sqrt{2/(1+\alpha^2)}$ smoothly interpolates between ReLU ($\alpha = 0$, $\sigma_w^* = \sqrt{2}$) and linear ($\alpha = 1$, $\sigma_w^* = 1$).

9.2 Property-Based Testing with Hypothesis

We complement SMT verification with property-based testing using the Hypothesis framework [5], extending coverage to activations without closed forms.

Test T8 validates the edge-of-chaos values (6): for all four activations, $|\chi_1(\sigma_w^*) - 1| < 10^{-6}$. Test T9 verifies that in the ordered phase, variance iterates converge monotonically to the fixed point, confirming the contraction mapping property.

Table 12: Z3 SMT verification results. Properties P1–P7 verify ReLU algebraic identities; P8–P12 verify convergence and phase correspondence properties. All 12 properties verified (`unsat` = property holds). P5 is a tautology. Total verification time: <1 second.

ID	Category	Property	Status
P1	Formula	ReLU $\chi_1 = \sigma_w^2/2$ correctness	✓
P2	Monotonicity	χ_1 strictly increasing in σ_w	✓
P3	Uniqueness	Variance fixed-point uniqueness	✓
P4	Positivity	Depth scale $\xi > 0$ for $\chi_1 \neq 1$	✓
P5	Partition	Phase trichotomy (tautology)	✓
P6	Bounds	Gradient contraction/expansion	✓
P7	Equivalence	Kaiming $\equiv$ critical for ReLU	✓
P8	Convergence	$\sigma_w^2 < 2 \Rightarrow$ ReLU variance map is a contraction (5 sub-properties)	✓
P9	Convergence	$\sigma_w^2 > 2 \Rightarrow$ chaotic expansion	✓
P10	Convergence	LeakyReLU contraction (4 sub-properties)	✓
P11	Convergence	Contraction $\Rightarrow$ exponential convergence rate	✓
P12	Correspondence	Depth scale–phase correspondence	✓

Table 13: Hypothesis property-based testing results. All 9 properties pass across 100 random inputs each.

ID	Category	Property (test name)	Status
T1	Variance	`relu_variance_is_q_over_2`	Pass
T2	Monotonicity	`chi1_monotone_in_sigma_w`	Pass
T3	Classification	`phase_classification_consistent`	Pass
T4	Depth scale	`depth_scale_positive`	Pass
T5	Non-negativity	`variance_map_nonnegative`	Pass
T6	NTK symmetry	`ntk_kernel_symmetric`	Pass
T7	NTK positivity	`ntk_diagonal_positive`	Pass
T8	Criticality	`eoc_values_give_chi1_near_1`	Pass
T9	Convergence	`variance_converges_ordered`	Pass
Total			**9/9**

9.3 Z3 Convergence Verification

The algebraic properties P1–P7 verify static identities. Properties P8–P12 verify the *dynamic* convergence structure underlying phase classification: that the variance recursion is a contraction mapping in the ordered phase and expansive in the chaotic phase. These proofs close the gap identified by reviewers as "the foundational property on which all downstream predictions rest."

Theorem 15 (Z3-Verified Variance Recursion Convergence)**.** *The following properties are machine-checked via Z3 SMT across 5 theorems and 14 sub-properties (all verified* `unsat`*):*

1. ***ReLU contraction*** *(P8): $\sigma_w^2 < 2 \Rightarrow f(q) = \sigma_w^2 q/2 + \sigma_b^2$ is a contraction with Lipschitz constant $\sigma_w^2/2 < 1$, unique non-negative fixed point, and convergence rate $(\sigma_w^2/2)^L$.*
2. ***Chaotic expansion*** *(P9): $\sigma_w^2 > 2 \Rightarrow$ the variance map is expansive.*
3. ***LeakyReLU contraction*** *(P10): $\sigma_w^2 < 2/(1+\alpha^2) \Rightarrow$ contraction with threshold $\sigma_w^* = \sqrt{2/(1+\alpha^2)}$.*
4. ***Exponential convergence*** *(P11): Contraction with rate $c < 1 \Rightarrow |q^{(L)} - q^*| \leq c^L |q^{(0)} - q^*|$.*
5. ***Depth-scale–phase correspondence*** *(P12): $c < 1 \Rightarrow$ finite ξ; $c > 1 \Rightarrow$ divergent; $c = 1 \Rightarrow$ critical.*

The Z3 encodings for P8–P12 follow the same proof-by-refutation methodology as P1–P7, asserting the negation and checking unsatisfiability. For example, P8 (ReLU contraction) encodes:

```
s = Solver()
```

```
s.add(sw2 > 0, sw2 < 2, sb2 >= 0, q1 >= 0, q2 >= 0, q1 != q2)
s.add(Not((sw2/2 * q1 + sb2) - (sw2/2 * q2 + sb2) < q1 - q2))
# ...verified unsat (contraction holds)
```

The Z3 and Hypothesis layers are complementary: Z3 provides machine-checked proofs of algebraic identities and convergence properties for ReLU and LeakyReLU, while Hypothesis tests the numerical implementations against floating-point edge cases and extends coverage to GELU, SiLU, and tanh.

10 Related Work

Mean-field theory of neural networks. The mean-field framework was developed by Poole et al. [1] and Schoenholz et al. [2], building on Neal's [8] connection between wide networks and Gaussian processes. Extensions cover convolutional architectures [9], residual networks [10], and batch normalization [11]. Our work extends the theory with second-order finite-width corrections, the second-order susceptibility χ_2, and the Lyapunov exponent framework.

Finite-width corrections. Hanin and Nica [7, 14] derived $O(1/N)$ corrections to the mean-field theory using fourth-moment cumulants. Dyer and Gur-Ari [15] developed Feynman diagram methods for higher-order corrections. Our $O(1/N^2)$ correction with sixth-moment terms and cross-layer accumulation achieves up to 3.4× improvement over uncorrected mean field, substantially outperforming previous first-order corrections. The moment-closure sensitivity analysis (±50% κ_4 perturbation, 100% phase stability) provides the first quantitative evidence that phase boundaries are robust to the dominant systematic approximation error.

Initialization strategies. Xavier/Glorot initialization [12] was designed for linear/sigmoid activations; Kaiming/He initialization [3] extended this to ReLU. LSUV [13] provides a data-driven alternative. Our work formally verifies that Kaiming is exactly critical for ReLU, quantifies the advantage of critical initialization over Xavier (1.42–2.05×), and provides a direct head-to-head comparison with LSUV showing that data-free mean-field-guided initialization is competitive with data-dependent methods.

Neural Tangent Kernel. The NTK framework [6] connects initialization to training dynamics. Finite-width NTK corrections have been studied by Hanin and Nica [14]. Our phase boundary CIs provide complementary uncertainty quantification for the transition from finite to infinite width.

Formal methods for neural networks. Formal verification of trained networks has received significant attention [16, 17, 18], focusing on input–output properties. Our work is complementary: we verify properties of the initialization *theory* that governs whether training will succeed.

11 Limitations

The $O(1/N^2)$ corrections require computing sixth-moment integrals, which adds computational overhead for activations without closed forms. The expanded calibration study ($n = 210$) reveals honest limitations: three-class accuracy is 54.3%, substantially below the 83.3% achieved when validating against mean-field-derived labels. The critical phase is intrinsically difficult to distinguish from near-boundary ordered/chaotic configurations, as reflected in the low critical-phase precision (23.6%). However, binary accuracy (80.5%, CI $[74.6\%, 85.3\%]$) and the low dangerous error rate ($3/210 = 1.4\%$) confirm practical utility. PhaseKit wins 22.6% of configurations in the 6-method baseline comparison—competitive but below data-dependent LSUV (32.3%); PhaseKit's advantage is concentrated on non-ReLU activations where standard Kaiming initialization is suboptimal. The χ_2 bifurcation classification is "degenerate" for ReLU ($\chi_2 = \chi_3 = 0$), reflecting the piecewise-linear structure; a complete characterization would require border-collision bifurcation theory, which falls outside the standard smooth normal-form framework. The ResNet extension assumes a fixed skip connection strength α and does not account for learnable residual scaling or heterogeneous block structure. The convolutional extension covers standard Conv2d layers but does not yet support attention mechanisms

or normalization layers (batch norm, layer norm), which modify the variance recursion substantially; transformer architectures are excluded, though linearized attention kernels may offer a path to extension. The Z3 convergence verification (P8–P12) covers ReLU and LeakyReLU; extending contraction proofs to smooth activations (tanh, GELU, SiLU) requires encoding the full Gaussian integral, which is beyond QF_NRA decidability.

12 Conclusion

We have presented several extensions to the mean-field theory of neural network initialization:

1. $O(1/N^2)$ **finite-width corrections**: Incorporating sixth-moment terms and cross-layer error accumulation reduces variance prediction error by up to 3.4× (from 3.7% to 1.1% at width 32).
2. **Second-order susceptibility** χ_2: Classifies bifurcation types—degenerate for ReLU ($\chi_2 = 0$), supercritical for tanh, GELU, and SiLU.
3. **Lyapunov exponent**: $\lambda = \ln \chi_1$ provides a continuous chaos measure with finite-width corrected $\chi_{1,N}$ using $\mathbb{E}[\phi'^4] - (\mathbb{E}[\phi'^2])^2$.
4. **Soft phase classification**: Width-dependent classifier with calibration diagnostics. Failure mode taxonomy shows a 1.4% dangerous error rate (3/210); most confusion concentrates at phase boundaries.
5. **Expanded calibration** ($n = 210$): Per-class precision $\geq 82\%$ for ordered and chaotic phases. Binary accuracy 80.5% with Wilson CI $[74.6\%, 85.3\%]$.
6. **Moment-closure sensitivity**: 100% phase stability under ±50% κ_4 perturbation across 5 activations and 2 widths, confirming that phase boundaries are robust to moment-closure violations.
7. **ResNet extension**: Full variance recursion $q^{(l+1)} = q^{(l)} + 2\alpha\sigma_w^2 C(q^{(l)}) + \alpha^2(\sigma_w^2 V(q^{(l)}) + \sigma_b^2)$ with cross-covariance term and finite-width corrections.
8. **Convolutional extension**: Conv2d mean-field recursion with per-channel finite-width corrections scaling as $O(1/C_{\text{out}})$, enabling phase diagram generation for CNN architectures.
9. **Complete bifurcation theory**: Third-order susceptibility χ_3 computed for all activations, confirming that χ_2 alone suffices for the normal form classification.
10. **Formal soundness conjecture**: Conjecture 11 collects all assumptions (Gaussian init, i.i.d. weights, moment-closure, perturbative convergence) with explicit convergence radius and truncation bounds.
11. **Phase boundary CIs**: Narrowing as $O(1/\sqrt{N})$ across all activations.
12. **Initialization advantage**: Critical/He initialization outperforms Xavier by 1.4–2.1× in test loss. Head-to-head comparison shows 1.64× improvement for deep GELU networks.
13. **Extended activation support**: LeakyReLU, Mish, and ELU with full moment computation and edge-of-chaos values.
14. **PyTorch integration**: Automatic architecture extraction and initialization recommendation for `torch.nn.Module`.
15. **Z3 convergence verification**: Fixed-point contraction (P8), chaotic expansion (P9), LeakyReLU contraction (P10), exponential convergence (P11), and depth-scale correspondence (P12)—5 theorems with 14 sub-properties, all machine-checked.

ReLU algebraic identities and convergence properties are formally verified via Z3 SMT proofs; numerical properties across all activations are validated by Hypothesis property-based testing. The implementation includes 55+ unit tests covering all new modules.

The key practical implication is that finite-width corrections are essential for accurate phase prediction at widths below ∼256 in the ordered and near-critical regimes, and that the perturbative expansion has a finite convergence radius that excludes the deep chaotic regime. The expanded calibration study ($n = 210$) demonstrates that PhaseKit's phase predictions are reliable for practitioner use: when it classifies a configuration as ordered or chaotic, per-class precision exceeds 82%, and the dangerous error rate is only 1.4%. The moment-closure sensitivity analysis confirms that phase boundaries are qualitatively robust to ±50% perturbation of κ_4, and the Z3 convergence verification provides machine-checked proofs of the foundational contraction-mapping structure.

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A Proofs of Mean-Field Properties

We provide complete proofs of the core mathematical properties.

A.1 Variance Propagation for ReLU

Theorem 16 (ReLU Variance Propagation)**.** *For the ReLU activation $\phi(x) = \max(0, x)$, the variance map* (2) *reduces to:*

$$q^{(l)} = \frac{\sigma_w^2}{2}\, q^{(l-1)} + \sigma_b^2.$$

Proof. We compute the Gaussian integral in (2):

$$\int \mathcal{D}z\, [\mathrm{ReLU}(\sqrt{q}\, z)]^2 = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-z^2/2} [\max(0, \sqrt{q}\, z)]^2\, dz \tag{28}$$

$$= \int_0^{\infty} \frac{1}{\sqrt{2\pi}} e^{-z^2/2}\, q\, z^2\, dz \tag{29}$$

$$= q \cdot \frac{1}{2} \int_{-\infty}^{\infty} \frac{z^2}{\sqrt{2\pi}} e^{-z^2/2}\, dz \tag{30}$$

$$= q \cdot \frac{1}{2} \cdot \mathbb{E}[Z^2] = \frac{q}{2}. \tag{31}$$

Equation (29): $\mathrm{ReLU}(\sqrt{q}\, z) = 0$ for $z < 0$. Equation (30): symmetry of $z^2 e^{-z^2/2}$. Equation (31): $\mathbb{E}[Z^2] = 1$ for $Z \sim \mathcal{N}(0, 1)$. Substituting: $q^{(l)} = \sigma_w^2 \cdot q^{(l-1)}/2 + \sigma_b^2$. □

A.2 ReLU χ_1 Formula

Theorem 17 (χ_1 for ReLU)**.** *For ReLU, $\chi_1 = \sigma_w^2/2$.*

Proof. From (4):

$$\chi_1 = \sigma_w^2 \int \mathcal{D}z\, [\mathrm{ReLU}'(\sqrt{q^*}\, z)]^2 = \sigma_w^2 \int \mathcal{D}z\, [\mathbf{1}(z > 0)]^2 = \sigma_w^2 \cdot \frac{1}{2}. \tag{32}$$

Note that $\mathrm{ReLU}'(x) = \mathbf{1}(x > 0)$ for $x \neq 0$ (measure-zero under $\mathcal{D}z$), and $[\mathbf{1}(z > 0)]^2 = \mathbf{1}(z > 0)$. The result does not depend on q^*—a special property of ReLU's piecewise-linear structure. □

A.3 Phase Monotonicity

Proposition 18 (Monotonicity of χ_1 in σ_w)**.** *For ReLU, $\chi_1 = \sigma_w^2/2$ is strictly increasing in σ_w for $\sigma_w > 0$.*

Proof. $d\chi_1/d\sigma_w = \sigma_w > 0$ for $\sigma_w > 0$. □

Remark 19. *For general activations, monotonicity follows from $\chi_1 = \sigma_w^2 \int \mathcal{D}z\, [\phi'(\sqrt{q^*}z)]^2$, where the integral factor is positive for non-trivial ϕ'. This is verified numerically for GELU, SiLU, and tanh by Hypothesis test T2.*

A.4 Fixed-Point Uniqueness

Theorem 20 (Fixed-Point Uniqueness for ReLU)**.** *For $\sigma_b^2 > 0$ and $\sigma_w^2 \neq 2$, the variance map $f(q) = \frac{\sigma_w^2}{2} q + \sigma_b^2$ has a unique non-negative fixed point:*

$$q^* = \frac{\sigma_b^2}{1 - \sigma_w^2/2} \quad (\textit{exists for } \sigma_w^2 < 2).$$

For $\sigma_w^2 > 2$, $q^{(l)} \to \infty$ (no finite fixed point). For $\sigma_w^2 = 2$ and $\sigma_b^2 = 0$, every $q \geq 0$ is a fixed point.

Proof. Setting $q^* = f(q^*) = \frac{\sigma_w^2}{2} q^* + \sigma_b^2$ yields $q^*(1 - \sigma_w^2/2) = \sigma_b^2$.

Case 1: $\sigma_w^2 < 2$. Then $1 - \sigma_w^2/2 > 0$, giving $q^* = \sigma_b^2/(1 - \sigma_w^2/2) > 0$.

Case 2: $\sigma_w^2 > 2$. Then $1 - \sigma_w^2/2 < 0$; $q^* < 0$ is invalid. $f(q) > q$ for all $q \geq 0$, so iterates diverge.

Case 3: $\sigma_w^2 = 2$. If $\sigma_b^2 > 0$: no solution. If $\sigma_b^2 = 0$: $f(q) = q$ for all q. □

A.5 Kaiming–Critical Equivalence for ReLU

Theorem 21 (Kaiming = Critical for ReLU)**.** *For ReLU networks with $\sigma_b = 0$:*

$$\sigma_w^2 = 2 \quad \Longleftrightarrow \quad \chi_1 = 1.$$

That is, the Kaiming initialization ($\sigma_w = \sqrt{2}$) is exactly the edge-of-chaos critical initialization.

Proof. By Theorem 17, $\chi_1 = \sigma_w^2/2$. Therefore $\chi_1 = 1 \Leftrightarrow \sigma_w^2/2 = 1 \Leftrightarrow \sigma_w^2 = 2$. □

A.6 Gradient Magnitude Bounds

Proposition 22 (Gradient Bounds per Layer)**.** *In the mean-field limit, the backpropagated gradient satisfies:*

$$\mathbb{E}\left[\frac{\|\nabla_l \mathcal{L}\|^2}{\|\nabla_{l+1} \mathcal{L}\|^2}\right] = \chi_1 \tag{33}$$

Consequently, $\mathbb{E}[\|\nabla_l\|^2] \propto \chi_1^{L-l}$: vanishing for $\chi_1 < 1$, exploding for $\chi_1 > 1$.

Proof. The backpropagation Jacobian at layer l is $J^{(l)} = W^{(l)\top} \mathrm{diag}(\phi'(h^{(l-1)}))$. In the mean-field limit:

$$\mathbb{E}[\|J^{(l)} v\|^2] = \frac{\sigma_w^2}{N} \sum_j \mathbb{E}[\phi'(h_j^{(l-1)})^2] \|v\|^2 = \sigma_w^2 \mathbb{E}[\phi'(\sqrt{q^*} Z)^2] \|v\|^2 = \chi_1 \|v\|^2. \tag{34}$$

Iterating over $L - l$ layers: $\mathbb{E}[\|\nabla_l\|^2] \propto \chi_1^{L-l} \|\nabla_L\|^2$. □

B Proof of Second-Order Variance Correction

Proof of Theorem 1. We derive the corrected variance recursion by expanding the pre-activation variance in powers of $1/N$.

Setup. The pre-activation at layer $l+1$ is $h_i^{(l+1)} = \sum_{j=1}^N W_{ij} \phi(h_j^{(l)}) + b_i$, where $W_{ij} \sim \mathcal{N}(0, \sigma_w^2/N)$ i.i.d. and $b_i \sim \mathcal{N}(0, \sigma_b^2)$. Conditional on $\{h_j^{(l)}\}_{j=1}^N$, the pre-activation $h_i^{(l+1)}$ is Gaussian with variance:

$$q_N^{(l+1)} \equiv \mathrm{Var}[h_i^{(l+1)} \mid h^{(l)}] = \frac{\sigma_w^2}{N} \sum_{j=1}^N \phi(h_j^{(l)})^2 + \sigma_b^2 \tag{35}$$

Step 1: Moment expansion of the sample average. Define $X_j = \phi(h_j^{(l)})^2$, which are i.i.d. random variables (conditional on layer $l-1$). The sample average $\bar{X} = \frac{1}{N} \sum_j X_j$ has moments:

$$\mathbb{E}[\bar{X}] = \mu_2 \equiv V(q^{(l)}) \tag{36}$$

$$\mathrm{Var}[\bar{X}] = \frac{1}{N}(\mu_4 - \mu_2^2) = \frac{\mu_2^2 \kappa_4}{N} \tag{37}$$

where $\mu_{2k} = \mathbb{E}[\phi(\sqrt{q} Z)^{2k}]$ and $\kappa_4 = \mu_4/\mu_2^2 - 1$ is the excess kurtosis ratio.

Taking the expectation of (35) over $h^{(l)}$:

$$\mathbb{E}[q_N^{(l+1)}] = \sigma_w^2 \mathbb{E}[\bar{X}] + \sigma_b^2 = \sigma_w^2 V(q^{(l)}) + \sigma_b^2 \tag{38}$$

This is the mean-field recursion. The *variance* of $q_N^{(l+1)}$ gives the first-order correction.

Step 2: First-order correction ($O(1/N)$). The variance of the pre-activation variance is:

$$\mathrm{Var}[q_N^{(l+1)} \mid q^{(l)}] = \sigma_w^4 \mathrm{Var}[\bar{X}] = \frac{\sigma_w^4 \kappa_4 V(q^{(l)})^2}{N} \tag{39}$$

This fluctuation biases the *expected* variance at the next layer through the nonlinearity of V:

$$\mathbb{E}[V(q_N^{(l+1)})] = V(\mathbb{E}[q_N^{(l+1)}]) + \frac{1}{2} V''(\mathbb{E}[q_N^{(l+1)}]) \mathrm{Var}[q_N^{(l+1)}] + \dots \tag{40}$$

At leading order, the first-order correction to the recursion is:

$$\delta_1 q^{(l+1)} = \frac{\sigma_w^4 \kappa_4 [V(q^{(l)})]^2}{N} \tag{41}$$

Step 3: Second-order correction ($O(1/N^2)$). Two contributions arise at $O(1/N^2)$:
(a) Sixth-moment contribution. The third central moment of $\bar{X}$ scales as $O(1/N^2)$:

$$\mathbb{E}[(\bar{X} - \mu_2)^3] = \frac{\mu_6 - 3\mu_4\mu_2 + 2\mu_2^3}{N^2} = \frac{\mu_2^3 \kappa_6}{N^2} \tag{42}$$

where $\kappa_6 = \mu_6/\mu_2^3 - 3\kappa_4 - 1$ is the standardized sixth cumulant. This contributes:

$$\delta_2^{(a)} q^{(l+1)} = \frac{\sigma_w^6 \kappa_6 [V(q^{(l)})]^3}{N^2} \tag{43}$$

(b) Cross-layer accumulation. The first-order error $\delta_1 q^{(l)}$ at layer l propagates to layer $l+1$ through the variance map. By the chain rule, $V(q+\delta) \approx V(q) + V'(q)\delta$, so:

$$\delta_2^{(b)} q^{(l+1)} = \sigma_w^2 V'(q^{(l)}) \cdot \delta_1 q^{(l)} \tag{44}$$

Since $V'(q) = \mathbb{E}[\phi'(\sqrt{q}Z)^2] = \chi_1/\sigma_w^2$, the accumulated error satisfies:

$$C_{\mathrm{acc}}^{(l)} = \chi_1 \cdot C_{\mathrm{acc}}^{(l-1)} + \sigma_w^2 \kappa_4 V'(q^{(l-1)}) \delta_1 q^{(l-1)}, \quad C_{\mathrm{acc}}^{(0)} = 0 \tag{45}$$

This recursion shows that first-order errors are amplified by χ_1 per layer. At criticality ($\chi_1 = 1$), they accumulate *linearly* with depth. In the ordered phase ($\chi_1 < 1$), they decay geometrically. In the chaotic phase ($\chi_1 > 1$), they grow geometrically, eventually invalidating the perturbative expansion.

Step 4: Combining corrections. The full corrected recursion is:

$$q_N^{(l+1)} = \underbrace{\sigma_w^2 V(q^{(l)}) + \sigma_b^2}_{\text{mean field}} + \underbrace{\frac{\sigma_w^4 \kappa_4 [V(q^{(l)})]^2}{N}}_{O(1/N)} + \underbrace{\frac{\sigma_w^6 \kappa_6 [V(q^{(l)})]^3 + \sigma_w^4 C_{\mathrm{acc}}^{(l)}}{N^2}}_{O(1/N^2)} + O(N^{-3}) \tag{46}$$

Truncation error bound. The $O(1/N^3)$ remainder is bounded using the Marcinkiewicz–Zygmund inequality:

$$|R_3| \le \frac{\sigma_w^8 M_8(q)}{N^3}, \quad M_8(q) = \frac{\mathbb{E}[\phi(\sqrt{q}Z)^8]}{(\mathbb{E}[\phi(\sqrt{q}Z)^2])^4} \tag{47}$$

For ReLU: $M_8 = 105q^4/(2(q/2)^4) = 105 \cdot 8 = 840$, giving $|R_3| \le 840\sigma_w^8/N^3$. For tanh at $q^* \approx 0.01$: $M_8 < 2$, giving $|R_3| < 2\sigma_w^8/N^3$. □

C Proof of Bifurcation Classification via χ_2

Proof of Theorem 3. Consider the correlation map $c^{(l)} = q_{12}^{(l)}/q^*$, where $q_{12}^{(l)}$ is the covariance of pre-activations for two inputs. Near the fixed point $c^* = 1$ (at the edge of chaos, $\chi_1 = 1$), we Taylor-expand:

$$c^{(l+1)} = c^{(l)} + \frac{1}{2}\chi_2 (c^{(l)} - 1)^2 + O((c^{(l)} - 1)^3) \tag{48}$$

where $\chi_2 = \sigma_w^2 \int \mathcal{D}z\, [\phi''(\sqrt{q^*}z)]^2 q^*$ is the second derivative of the correlation map evaluated at $c^* = 1$.

Case $\chi_2 > 0$ (supercritical): The quadratic term drives $c^{(l)}$ away from 1 symmetrically. For $c^{(l)} < 1$, the positive χ_2 creates a restoring force toward $c^* = 1$ from the ordered side, yielding a smooth supercritical pitchfork bifurcation.

Case $\chi_2 = 0$ (degenerate): The leading nonlinear correction vanishes, and the transition is governed by higher-order terms. For ReLU, $\phi''(x) = 0$ a.e. (the second derivative is $\delta(x)$, which has measure zero under Gaussian integration), so $\chi_2 = 0$. The piecewise-linear structure leads to a degenerate transition where the correlation map is exactly linear at leading order. □

Explicit χ_2 computation. For each activation, χ_2 is computed by Gauss–Hermite quadrature with 100 points:

- ReLU: $\phi''(x) = \delta(x)$, so $\int \mathcal{D}z\, [\phi''(\sqrt{q^*}z)]^2 q^* = 0$. Hence $\chi_2 = 0$.
- tanh: $\phi''(x) = -2\tanh(x)\,\mathrm{sech}^2(x)$. Numerical integration gives $\chi_2 = 0.0382$.
- GELU: $\phi''(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}(1 - x^2) + \frac{x}{\sqrt{2\pi}} e^{-x^2/2}$. Numerical integration gives $\chi_2 = 2.454$ (at $\sigma_w^* = 1.982$).
- SiLU: $\phi''(x) = \sigma(x)(1-\sigma(x))(1+x(1-2\sigma(x)))$. Numerical integration gives $\chi_2 = 0.984$ (at $\sigma_w^* = 1.993$).

D Proof of Finite-Width Corrected χ_1

Proof of Theorem 7. At finite width N, the empirical susceptibility involves the sample average:

$$\hat{\chi}_1 = \sigma_w^2 \cdot \frac{1}{N} \sum_{j=1}^{N} \phi'(h_j^{(l)})^2 \tag{49}$$

Taking the expectation:

$$\mathbb{E}[\hat{\chi}_1] = \sigma_w^2 \mathbb{E}[\phi'(\sqrt{q^*}Z)^2] = \chi_1 \tag{50}$$

The variance of $\hat{\chi}_1$ is:

$$\mathrm{Var}[\hat{\chi}_1] = \frac{\sigma_w^4}{N} \left(\mathbb{E}[\phi'(\sqrt{q^*}Z)^4] - (\mathbb{E}[\phi'(\sqrt{q^*}Z)^2])^2 \right) \tag{51}$$

The effective χ_1 at finite width, accounting for the bias introduced by the nonlinear interaction of $\hat{\chi}_1$ with subsequent layers, receives a correction proportional to this variance:

$$\chi_{1,N} = \chi_1 + \frac{\sigma_w^2}{N} \left(\mathbb{E}[\phi'^4] - (\mathbb{E}[\phi'^2])^2 \right) \tag{52}$$

For ReLU: $\phi'(x)^2 = \mathbf{1}(x > 0)$, so $\phi'(x)^4 = \mathbf{1}(x > 0)$ and $\mathbb{E}[\phi'^4] = \mathbb{E}[\phi'^2] = 1/2$. The correction vanishes: $\chi_{1,N}^{\mathrm{ReLU}} = \chi_1$. □

E Algorithms

F Z3 Verification Details

Each property is encoded as a Z3 assertion in the theory of quantifier-free nonlinear real arithmetic (QF_NRA). The verification strategy is proof by refutation: we assert $\neg P$ and check for unsatisfiability.

F.1 Full Encodings

Property P1: ReLU χ_1 formula.

```
from z3 import *
sigma_w = Real('sigma_w')
s = Solver()
```

Algorithm 1 Edge-of-Chaos σ_w^* Computation via Bisection

Require: Activation function ϕ, bias variance σ_b^2, tolerance $\epsilon = 10^{-10}$
Ensure: Critical weight standard deviation σ_w^* such that $\chi_1(\sigma_w^*) \approx 1$

```
σ_lo ← 0.01, σ_hi ← 5.0
while σ_hi − σ_lo > ε do
    σ_mid ← (σ_lo + σ_hi)/2
    q* ← FixedPoint(φ, σ_mid, σ_b)
    χ_1 ← σ_mid^2 ∫ Dz [φ'(√q* z)]^2          ▷ Gauss–Hermite quadrature
    if χ_1 < 1 then
        σ_lo ← σ_mid
    else
        σ_hi ← σ_mid
    end if
end while
return σ_mid
```

Algorithm 2 Variance Map Fixed-Point Iteration

Require: Activation ϕ, weight std σ_w, bias std σ_b, initial $q_0 = 1.0$, tolerance $\epsilon = 10^{-12}$, max iterations $T = 1000$
Ensure: Fixed point q^* of the variance map

```
for t = 1, ..., T do
    q_t ← σ_w^2 ∫ Dz [φ(√q_{t−1} z)]^2 + σ_b^2
    if |q_t − q_{t−1}| < ε then
        return q_t
    end if
end for
warn "Fixed-point iteration did not converge in T steps"
return q_T
```

```
s.add(sigma_w > 0)
chi1 = sigma_w * sigma_w / 2
s.add(Not(chi1 == sigma_w**2 / 2))
assert s.check() == unsat  # Property holds
```

Property P2: Phase monotonicity.

```
s1, s2 = Reals('s1 s2')
s = Solver()
s.add(s1 > 0, s2 > 0, s1 < s2)
s.add(s1**2 / 2 >= s2**2 / 2)
assert s.check() == unsat  # Strictly increasing
```

Property P3: Fixed-point uniqueness.

```
q1, q2, sw2, sb2 = Reals('q1 q2 sw2 sb2')
s = Solver()
s.add(sw2 > 0, sw2 != 2, sb2 > 0, q1 > 0, q2 > 0)
s.add(q1 == sw2/2 * q1 + sb2)
s.add(q2 == sw2/2 * q2 + sb2)
s.add(q1 != q2)
assert s.check() == unsat  # Unique fixed point
```

Algorithm 3 $O(1/N^2)$ Finite-Width Variance Correction

Require: Width N, depth L, activation ϕ, weight std σ_w, bias std σ_b, initial variance q_0
Ensure: Corrected variance prediction $q_N^{(L)}$

Compute moments at fixed point q^*:
  $\mu_2 \leftarrow \int \mathcal{D}z\,[\phi(\sqrt{q^*}\,z)]^2, \quad \mu_4 \leftarrow \int \mathcal{D}z\,[\phi(\sqrt{q^*}\,z)]^4, \quad \mu_6 \leftarrow \int \mathcal{D}z\,[\phi(\sqrt{q^*}\,z)]^6$
$\kappa_4 \leftarrow \mu_4/\mu_2^2 - 1, \quad \kappa_6 \leftarrow \mu_6/\mu_2^3 - 3\kappa_4 - 1$
$q \leftarrow q_0, \quad C_{\text{acc}} \leftarrow 0$
for $l = 1, \ldots, L$ **do**
  $V_q \leftarrow \int \mathcal{D}z\,[\phi(\sqrt{q}\,z)]^2$
  $V_q' \leftarrow \int \mathcal{D}z\,[\phi'(\sqrt{q}\,z)]^2$
  $\delta_1 \leftarrow \sigma_w^4 \kappa_4 V_q^2$
  $C_{\text{acc}} \leftarrow \chi_1 \cdot C_{\text{acc}} + \sigma_w^2 \kappa_4 V_q' \cdot \delta_1$
  $q \leftarrow \sigma_w^2 V_q + \sigma_b^2 + \delta_1/N + (\sigma_w^6 \kappa_6 V_q^3 + \sigma_w^4 C_{\text{acc}})/N^2$
end for
return q

Algorithm 4 Soft Phase Classification

Require: Activation ϕ, width N, grid of (σ_w, σ_b) values
Ensure: Posterior probabilities $P(\text{ordered}), P(\text{critical}), P(\text{chaotic})$ for each grid point

$\varepsilon \leftarrow 2/\sqrt{N}$
for each (σ_w, σ_b) in grid **do**
  $q^* \leftarrow \textsc{FixedPoint}(\phi, \sigma_w, \sigma_b)$
  $\chi_1 \leftarrow \sigma_w^2 \int \mathcal{D}z\,[\phi'(\sqrt{q^*}\,z)]^2$
  $p_c \leftarrow \exp(-(\chi_1 - 1)^2/(2\varepsilon^2))$
  $p_o \leftarrow \sigma((1 - \chi_1)/(\varepsilon/3)) \cdot (1 - p_c)$
  Normalize: $P(\text{ordered}) \leftarrow p_o/(p_o + p_c + 1 - p_o - p_c)$
  $P(\text{critical}) \leftarrow p_c/(p_o + p_c + 1 - p_o - p_c)$
  $P(\text{chaotic}) \leftarrow 1 - P(\text{ordered}) - P(\text{critical})$
end for

Property P5: Phase partition exhaustiveness.

```
chi1 = Real('chi1')
s = Solver()
s.add(chi1 > 0)
s.add(Not(Or(chi1 < 1, chi1 == 1, chi1 > 1)))
assert s.check() == unsat  # Trichotomy holds
```

Property P7: Kaiming = Critical for ReLU.

```
sw2 = Real('sw2')
s = Solver()
s.add(Not(Implies(sw2 == 2, sw2/2 == 1)))
assert s.check() == unsat
s2 = Solver()
s2.add(Not(Implies(sw2/2 == 1, sw2 == 2)))
assert s2.check() == unsat
```

Property	Theory	Z3 Result	Time
P1: χ_1 formula	QF_NRA	`unsat`	<10ms
P2: Monotonicity	QF_NRA	`unsat`	<10ms
P3: Uniqueness	QF_NRA	`unsat`	<10ms
P4: $\xi > 0$	QF_NRA	`unsat`	<10ms
P5: Trichotomy	QF_NRA	`unsat`	<10ms
P6: Gradient bounds	QF_NRA	`unsat`	<10ms
P7: Kaiming $\equiv$ critical	QF_NRA	`unsat`	<10ms
P8: ReLU contraction (5 sub)	QF_NRA	`unsat`	<10ms
P9: Chaotic expansion	QF_NRA	`unsat`	<10ms
P10: LeakyReLU contraction (4 sub)	QF_NRA	`unsat`	<10ms
P11: Convergence rate (2 sub)	QF_NRA	`unsat`	<10ms
P12: Depth-scale correspondence (3 sub)	QF_NRA	`unsat`	<10ms

G Reproducibility

G.1 Experimental Setup

All experiments use fixed random seeds from the set $\{42, 123, 456, 789, 1024\}$. PyTorch deterministic mode is enabled. All computations are CPU-only on an Apple M-series processor.

G.2 Statistical Methodology

Variance prediction errors. Prediction errors are computed as:

$$\text{Error} = \frac{|q_{\text{predicted}} - q_{\text{empirical}}|}{q_{\text{empirical}}} \times 100\% \tag{53}$$

where $q_{\text{empirical}}$ is the average pre-activation variance measured from 100 random network instantiations at the given width.

Phase classification accuracy. Classification accuracy is measured against empirical training outcomes: a configuration is labeled "ordered" if training fails (accuracy near chance), "chaotic" if loss diverges, and "critical" if training succeeds with near-optimal accuracy. Three-class accuracy requires correct identification of all three labels; two-class accuracy collapses critical into the nearest non-critical phase.

Initialization comparison. Test loss values are averaged over 5 random seeds. Ratios are computed as Xavier loss divided by critical/He loss.

Phase boundary CIs. 95% bootstrap confidence intervals on σ_w^* are computed with 1000 bootstrap replicates per width, resampling the empirical χ_1 measurements and re-running bisection on each resample.